

Raj Singh

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— Portfolio — GitHub — LinkedIn

Professional Summary

Full-Stack Developer with experience building scalable web applications and AI-powered systems using MERN Stack, TypeScript, LangGraph, LangChain, Pinecone, Docker & kubernetes, CI/CD pipeline and Retrieval-Augmented Generation (RAG). Built and deployed multiple production-ready applications, contributed to open-source programs, and continuously explore modern AI technologies and software engineering practices.

Technical Skills

Languages: JavaScript, TypeScript, HTML, CSS, C

Frontend: React.js, Next.js, Redux Toolkit, Tailwind CSS, Responsive Design

Backend: Node.js, Express.js, REST APIs, JWT Authentication, Nodemailer

Databases: MongoDB, Redis, Pinecone Vector Database

AI/GenAI: LangGraph, LangChain, RAG Systems, AI Agents, Tool Calling, Prompt Engineering, Multi-LLM Routing

Tools: Git, GitHub, Postman, VS Code, Vercel, Netlify, Render

Projects

Perplex AI

live demo

- Built a Perplexity-inspired AI assistant using LangGraph workflows, Pinecone memory, and RAG architecture.
- Integrated Tavily Search API for real-time information retrieval.
- Implemented intelligent routing across Gemini, Mistral, and Groq models.

Moodify

Live Demo

- Developed a MERN application using MediaPipe for real-time emotion detection and music recommendations.
- Implemented JWT authentication, email verification, and role-based access control.

Snitch

Live Demo

- Built a full-stack social platform with real-time posts, user interactions, and role-based access control.
- Implemented JWT authentication, Redux state management, media uploads, and optimized API performance.

AI Battle Arena

Live Demo

- Developed an AI comparison platform enabling side-by-side battles between multiple LLMs with unified responses.
- Integrated multi-model routing, prompt orchestration, conversation history, and real-time result streaming.

Projects & Development

- Built and deployed multiple full-stack applications using React, Next.js, Node.js, Express, and MongoDB.
- Actively explored AI technologies including RAG, LangGraph, multi-agent systems, and LLM integrations. Consistently maintained personal projects with version control, documentation, and production deployments.

Education

Bachelor of Technology (B.Tech) in Computer Science and Engineering (CSE)

Mangalmay Institute of Engineering and Technology College

2024-2028

Certifications

- Postman API Beginner Learning Path – Postman Academy
- Prompt Engineering Bootcamp – LetsUpgrade / NSDC
- Git & GitHub Bootcamp – LetsUpgrade / NSDC
- Communication Skills Certification – Microsoft & AISECT

Achievements

- Secured 1st Position in an Internal College Hackathon for developing and presenting an innovative software solution.
- Developed and deployed AI-powered and full-stack applications using MERN, LangGraph, and RAG architectures.
- Maintained 140+ GitHub contributions and consistently developed personal projects using Git and GitHub.